

Alireza (Aaron) Imani

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SUMMARY

Alireza (Aaron) Imani is a PhD candidate in Software Engineering at the University of California, Irvine. He holds a Masters degree in Software Engineering from the University of Calgary. With five years of research experience, he has published six peer-reviewed papers in prestigious international conferences, including the International Conference on Software Engineering (ICSE) and the International Conference on Mining Software Repositories (MSR). He also serves as a reviewer for top-tier journals such as IEEE Transactions on Reliability and ACM Transactions on Software Engineering and Methodology (TOSEM). His current research focuses on understanding the underlying mechanisms of large language models in downstream software engineering tasks, from a mechanistic interpretability perspective.

EDUCATION

- **University of California, Irvine** Expected 06 / 2027
Ph.D. in Software Engineering Irvine, USA
 - GPA: 4.00 / 4.00
 - Dissertation: Interpreting Large Language Models in Code Intelligence Tasks
- **University of Calgary** 08 / 2023
M.Sc. in Software Engineering Calgary, Canada
 - GPA: 4.00 / 4.00
 - Dissertation: [Effective Control System Framework Selection through Checklist-based Software Quality Evaluation](#)
- **Ferdowsi University of Mashhad** 09 / 2020
B.Sc. in Computer Engineering Mashhad, Iran
 - Final Project: Workflow Scheduling Using Artificial Bee Colony Optimization

EXPERIENCE

- **University of California, Irvine - STAIRS Lab**  09/2023 – Present
Research Assistant Irvine, USA
 - Published three peer-reviewed papers.
 - Designed and evaluated innovative LLM-based approaches for code-related tasks, such as commit message generation, enhancing developer productivity.
 - Investigated the interpretability of LLMs in code understanding
- **University of Calgary - SEPE Lab**  01/2021 – 08/2023
Research Assistant Calgary, Canada
 - Published three peer-reviewed papers.
 - Explored methods to assess and improve the quality of open-source software, contributing to community-driven development practices.
 - Studied the challenges of requirements engineering in multidisciplinary teams
- **Sessional Instructor** Calgary, Canada
University of Calgary 08/2022 – 12/2022
 - Lectured a graduate-level course, “Programming Fundamentals for Data Engineers”, which covers the fundamentals of Python programming, including Basic data structures and algorithms; Loops and iterations; Files and I/O, Functions, Classes, Modules, and Packages; Strings and text manipulation; Data wrangling; Network and Web programming, and Data visualization.

- Contributed to the ARTTA-4 project, focusing on control system enhancements.
- Developed interconnected device servers in C++ and Python using the Tango Controls framework for the control system managing advanced radio telescopes at Dominion Radio Astrophysical Observatory.
- Facilitated seamless communication with hardware devices, such as a Moxa serial device server and a LabJack T4 device, by developing robust software interfaces.

PUBLICATIONS

C=CONFERENCE

- [C.6] **Aaron Imani**, Mohammad Moshirpour, and Iftekhar Ahmed (2025). **Inside Out: Uncovering How Comment Internalization Steers LLMs for Better or Worse**. In *2026 IEEE/ACM 48th International Conference on Software Engineering (ICSE)*.
- [C.5] **Aaron Imani**, Iftekhar Ahmed, and Mohammad Moshirpour. (2025) **Context Conquers Parameters: Outperforming Proprietary Llm in Commit Message Generation**. In *2025 IEEE/ACM 47th International Conference on Software Engineering (ICSE)*, pp. 1844-1856. IEEE. Ottawa, ON, Canada. DOI: 10.1109/ICSE55347.2025.00048
- [C.4] Mahan Tafreshipour, **Aaron Imani**, Eric Huang, Eduardo S. d. Almeida, Thomas Zimmermann and Iftekhar Ahmed (2025) **Prompting in the Wild: An Empirical Study of Prompt Evolution in Software Repositories**. In *IEEE/ACM 22nd International Conference on Mining Software Repositories (MSR)*, pp. 1844-1856. IEEE. Ottawa, ON, Canada. DOI: 10.1109/ICSE55347.2025.00048
- [C.3] Ali Salmani, **Alireza Imani**, Majid Bahrehvar, Linda Duffett-Leger, and Mohammad Moshirpour (2022) **A Data-Centric Approach to Evaluate Requirements Engineering in Multidisciplinary Projects**. In *2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 903-908. IEEE. Prague, Czech Republic. DOI: 10.1109/SMC53654.2022.9945270
- [C.2] Ali Salmani, **Alireza Imani**, Majid Bahrehvar, Linda Duffett-Leger, and Mohammad Moshirpour (2022) **An Intelligent Methodology to Enhance Requirements Engineering in Multidisciplinary Projects**. In *2022 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*, pp. 452-457. IEEE. Halifax, NS, Canada. DOI: 10.1109/CCECE49351.2022.9918286
- [C.1] **Alireza Imani**, Mohammad Moshirpour, and Leonid Belostotski (2021) **Checklist-based Software Quality Evaluation of Tango Controls**. In *2021 International Conference on Information Systems and Advanced Technologies (ICISAT)*, pp. 1-7. IEEE. Tebessa, Algeria. DOI: 10.1109/ICISAT54145.2021.9678411

TEACHING

Lecturer

- University of Calgary, **Programming Fundamentals for Data Engineers** 08/2022 – 12/2022

Teaching Assistant

- University of California, Irvine, I&C SCI 32, **Programming with Software Libraries** 03/2025 – 06/2025
- University of California, Irvine, I&C SCI 32, **Programming with Software Libraries** 01/2025 – 03/2025
- University of California, Irvine, SWE 240P, **Data Structures** 09/2024 – 12/2024
- University of California, Irvine, SWE 241P, **Algorithms** 09/2024 – 12/2024
- University of California, Irvine, I&C SCI 45J, **Programming in Java as a Second Language** 08/2024 – 09/2024
- University of California, Irvine, I&C SCI 32, **Programming with Software Libraries** 03/2024 – 06/2024
- University of California, Irvine, I&C SCI 32, **Programming with Software Libraries** 01/2024 – 03/2024
- University of California, Irvine, IN4MATX 117, **Project in Software System Design** 09/2023 – 12/2023
- University of Calgary, SENG401, **Software Architecture** 01/2023 – 04/2022
- University of Calgary, ENSF480, **Principles of Software Design** 08/2021 – 12/2021
- University of Calgary, ENSF609, **Team Design Project in Software Engineering** 01/2021 – 04/2021
- Ferdowsi University of Mashhad, **Data Structures** 03/2019 – 06/2019

PROJECTS

• **OMEGA: State-of-the-art Automated Commit Message Generation Tool**

03/2024 - 08/2024

Tools: Python, Langchain, vllm



- Developed OMEGA, a state-of-the-art tool for commit message generation that prioritizes data privacy, reduces carbon footprint, and lowers deployment costs
- Enhanced privacy by eliminating reliance on proprietary large language models, ensuring secure and ethical AI adoption.
- Optimized sustainability by adopting a 4-bit quantized open-source large language model runnable on local GPUs with only 8GB of VRAM.
- Proposed FIDEX, a novel diff augmentation technique that improved LLM performance on understanding code changes by 20%
- Adapted Multi-intent Method Summarization to support change-centric software engineering tasks, enabling better summarization of code edits

SERVICE ACTIVITIES

Reviewer

- ACM Transactions on Software Engineering and Methodology 11/2024 – Present
- IEEE Transactions on Reliability 10/2024 – Present

Volunteer

- Chair, IEEE Southern Alberta Computer Society Chapter 01/2023 – 08/2023
- Vice Chair, IEEE Southern Alberta Computer Society Chapter 04/2021 – 12/2022

Student Mentor

- Mentor, MedTech Innovation Hackathon 10/2024
- Mentor, ICS Honors Program 01/2024 – 07/2024

SKILLS

- **Research Skills:** Natural Language Processing, Large Language Models, Fine-tuning, Prompt Engineering, Model Training, Machine Learning, Neural Network, Dataset Curation, Dataset Labeling, Model Evaluation, Deep Learning, RNN, LSTM, Literature Review, Empirical Research, Interview / Survey Design, Qualitative Analysis, Statistical Analysis, Human Subjects Research, Statistics, IoT
- **Programming Languages:** Python, C, C++, Java, Swift, JavaScript
- **Web Technologies:** ReactJS, HTML5, CSS, Django, REST API
- **Database Systems:** MySQL, SQLite, Dynamo DB, Firebase
- **Data Science & Machine Learning:** PyTorch, SGLang, Langchain, Scikit-learn, NLTK, transformers
- **Cloud Technologies:** AWS, Heroku, Firebase
- **DevOps & Version Control:** Git, Docker, Conda

HONORS AND AWARDS

• **Chair's Award**

09/2023

Donald Bren School of Information and Computer Sciences

- Received upon admission to the PhD program at the University of California, Irvine
- Granted in recognition of exceptional potential to make significant research contributions to the Informatics community

• **Graduate Student Productivity Reward**

09/2022

University of Calgary

- Granted to graduate students who published at least one paper in a peer-reviewed conference

• AI Competition Achievement

Sharif Artificial Intelligence Challenge

- Designed an innovative AI solution for a real-time strategic game
- Ranked among the top 8 teams in the Main Game track out of over 1,000 participants
- Achieved 2nd place in the Mini Game track

03/2019



REFERENCES

1. Prof. Iftekhar Ahmed

Associate Professor, Donald Bren School of Information and Computer Sciences

University of California, Irvine

Email: iftekha@uci.edu

Relationship: PhD Advisor

2. Prof. Thomas Zimmermann

Chancellors Professor and Bren Chair, Donald Bren School of Information and Computer Sciences

University of California, Irvine

Email: tzimmer@uci.edu

Relationship: Faculty Collaborator